Rithvika Sapalya

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Education

- **Rutgers, New Brunswick** — New Brunswick, NJ
  Master of Science in Computer Science; GPA: 3.83 — *Sep. 2025 – May. 2027*
- **Thakur College of Engineering and Technology** — Mumbai, India
  Bachelor of Engineering in Electronics and Conputer Science; GPA: 3.82 (9.54/10.0) — *Dec. 2021 – June. 2025*

Experience

- **Full-Stack Document Scanner SDK** — Mumbai, India
  Full-Stack Development Intern — *Dec 2024 - Mar 2025*
  - Directed the integration of Canon EDSDK to programmatically control camera settings, trigger captures, and transfer images directly from connected devices to the backend pipeline, eliminating manual intervention in the scanning workflow.
  - Architected an image preprocessing pipeline using **ImageMagick** for resolution tuning, noise reduction, and binarization, driving a 17% improvement in OCR accuracy by ensuring clean, high-contrast input before text extraction.
  - Spearheaded implementation of multi-model **Tesseract OCR** with domain-specific trained models, standard English (eng), MICR E-13B (e13b) for bank cheque and financial document processing, and MICR font support, extending the system's applicability across general, banking, and enterprise document workflows.

Projects

- **The Reader's Circle** — Java Full Stack
  Java, Spring Boot, Spring Cloud, MySQL, React, TypeScript — *Github*
  - Architected and delivered a full-stack social book cataloguing platform using a **7-service microservice architecture**, separating concerns cleanly across independent, deployable Spring Boot applications.
  - Engineered a Spring Cloud Gateway as the unified API entry point with dynamic service discovery via Eureka, routing all client traffic to downstream services without hardcoded URLs, and exposed aggregated Swagger/OpenAPI docs across all services through a single gateway endpoint.
  - Built JWT-based stateless authentication with BCrypt password hashing and per-service JwtFilter middleware, enforcing @PreAuthorize role-based access control (USER / AUTHOR / ADMIN roles) across all secured endpoints.
  - Designed a MySQL relational schema across 9 tables (user, book, author, post, review, comment, wishlist, auth, and sequences) with genre arrays, buy-site links, and review ID associations stored against book records.
  - Developed a React + TypeScript frontend with protected routes, JWT-aware Axios interceptors, and dedicated pages for Home, Genres, Authors, Books, Posts, Wishlist, Search, Admin, and Settings — deployed via Netlify with SPA redirect support.
- **Voice Cloning for ALS Patients** — Machine Learning
  Python, PyTorch, SpeechT5, LSTM, HiFi-GAN, FastAPI, TORGO — *Github*
  - Designed a **4-stage** fine-tuning pipeline on the TORGO dysarthric corpus, 3-layer LSTM speaker encoder (GE2E loss), SpeechT5 postnet refinement, and anchor-regularised per-speaker x-vector adaptation, lifting baseline dysarthric speaker similarity from **0.55 to 0.76–0.82** on held-out speakers.
  - Engineered a **Newton-style iterative momentum refinement** at inference: each pass corrects the speaker embedding toward the target, generating N candidates and selecting the best, achieving **+34–48% relative improvement** in speaker similarity at 3–5× inference cost.
  - Achieved **0.829** mean speaker similarity on control and 0.778 on dysarthric speakers across all 15 TORGO speakers, with held-out speakers generalising within 0.03 of training averages.
  - Built a FastAPI backend with a drag-and-drop cloning UI and a 15-speaker evaluation dashboard with live audio samples and per-speaker breakdowns.

Programming Skills

- **Languages**: Java, Python, JavaScript, C++, C, HTML **Frameworks**: Java Full Stack, SpringBoot, TensorFlow, Scikit-Learn, SQL, MongoDB, MERN Stack **Areas of Interest**: Full Stack Development, Speech Synthesis, Machine Learning, Computer Vision **Certificates**: Machine Learning Specialization - Coursera, Geodata Processing using Python - ISRO